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Research on collision avoidance route planning based on A-star algorithm

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Abstract

The increasing complexity of maritime traffic necessitates advanced collision avoidance systems for autonomous navigation. Traditional path planning algorithms often struggle with dynamic obstacle prediction and adherence to navigational regulations. To address these challenges, this study introduces a collision avoidance framework that integrates the Obstacle Zone by Target (OZT) method with an improved A-star algorithm. This paper proposes a combination of the OZT method and an improved A-star algorithm, which incorporates a decision-making module and segmented path planning to generate COLREGs-compliant avoidance routes, optimizing route safety and smoothness. The effectiveness of this system was verified through simulations, and the results demonstrate the system's ability to generate collision-free paths in multi-ship scenarios while considering and complying with COLREGs. Compared to the traditional A-star algorithm, the proposed approach reduced path length by an average of 5.92% and decreased turning points by 79.44% in give-way situations, significantly improving navigation efficiency and path smoothness.

Keywords**:** *ship collision avoidance, route planning, OZT, A-star algorithm, maritime safety*

1. INTRODUCTION

1.1 Background

With the rapid growth of global maritime trade, high-density traffic in congested waterways has become increasingly common. This surge in vessel numbers not only intensifies the risk of collisions but also places considerable pressure on navigators and vessel traffic service operators. Although various technological advancements and COLREGs (The International Regulations for Preventing Collisions at Sea) have been introduced to enhance situation awareness and reduce accidents, human errors remain a major contributing factor to collisions and near-miss incidents, resulting in significant economic and operational losses[1)].

In response, the development of autonomous and semi-autonomous ships has garnered considerable attention worldwide. By removing or reducing reliance on human operators, such vessels have the potential to minimize collisions, thereby mitigating economic losses and enhancing overall maritime safety. Among the critical components of autonomous navigation systems, collision avoidance route planning plays a vital role in generating collision-free paths.

1.2 Related works

In complex maritime environments, effective collision avoidance requires accurate risk assessment to develop strategies that safely navigate around obstacles. This paper reviews both collision risk assessment and path planning methods for ship navigation.

1.2.1 Development of Collision Risk Assessment Methods

Early approaches relied primarily on the distance to a closest point of approach (DCPA) and time to a closest point of approach (TCPA) calculations to evaluate collision risk, as discussed by Kouzuki and Hasegawa who integrated these parameters into a fuzzy controller to determine optimal timing for collision avoidance actions[2)]. Szlapczynski redefined collision risk calculation by integrating ship domain theory with DCPA/TCPA parameters[3)].

The Obstacle Zone by Target (OZT) method represents a significant advancement in collision risk modeling. Initially introduced by Imazu[4)], OZT estimates a potential collision area based on ship dynamics, with its radius representing the safe passing distance. This method creates a graphical representation of areas where collision would occur if the own ship entered, offering a more intuitive understanding of risk. Fukuto and Imazu[5)] further developed this concept into a collision alarm algorithm that demonstrates superior performance in congested waterways compared to traditional DCPA/TCPA methods.

However, these conventional methods utilize static parameters for risk assessment, making them unable to adapt to the complex vessel interactions in high-density traffic

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environments. Such approaches struggle to accurately evaluate comprehensive risks when multiple vessels are simultaneously navigating in confined waterways, where interactions become increasingly dynamic and unpredictable.

1.2.2 Evolution of Ship Collision Avoidance Planning

Ship collision avoidance planning has evolved from manual chart-based methods to computational approaches, accelerated by ECDIS and ARPA systems. Traditional path planning methods like artificial potential field approaches gained popularity due to their computational efficiency, with Huang et al.[6] incorporating ship maneuverability characteristics to create more realistic avoidance behaviors.

Search-based methods have also advanced significantly. Xuan[7] demonstrated effective A-star algorithm implementations for unmanned vessels, balancing computational efficiency with path optimization.

Neural network and reinforcement learning approaches show promise, with Wang et al.[8] demonstrating autonomous ship obstacle avoidance through self-learning. However, these methods require extensive training data and computational resources, limiting real-time applicability. Sawada et al.[9], achieved good results in trajectory prediction and evasion planning, but these approaches lack generalizability across maritime environments and their black-box nature complicates COLREGs compliance verification and decision explanation, hindering practical implementation.

1.3 Research Gaps and Proposed Contributions

Based on our literature review, we identified three significant gaps in current research:

1. Risk representation gap: Most existing approaches use static parameters that fail to adapt to dynamic multi-ship interactions in congested waterways.
2. COLREGs compliance gap: Current algorithms struggle to consistently incorporate maritime regulations across different encounter scenarios.
3. Computational efficiency gap: Advanced methods like reinforcement learning lack the real-time performance necessary for practical implementation.

Our research addresses these gaps through an integrated framework combining OZT methodology with an improved A-star algorithm. This approach provides enhanced obstacle representation through high-resolution mapping of moving vessels, COLREGs-compliant decision-making with appropriate evasion options, and efficient path planning that balances safety, smoothness, and regulatory compliance.

By integrating these elements, our method achieves superior performance in dense traffic scenarios while maintaining computational efficiency for real-time applications. We validate our approach using both standard test scenarios and real-world traffic data from Tokyo Bay.

1.4 Paper Organization

The remainder of this paper is structured as follows.

Chapter 2 introduces the theoretical foundations of the traditional A-star algorithm, OZT method and the improved A-star algorithm, including its dynamic node expansion and OZT-based cost function.

Chapter 3 describes the simulation setup and experimental design within Tokyo Bay, detailing performance metrics and presenting the results of both single-ship and multi-ship collision avoidance tests.

Chapter 4 offers a detailed discussion of the results in terms of COLREGs compliance, system performance and the limitations of the proposed method.

Chapter 5 concludes the paper by summarizing the main contributions.

2. METHODS

2.1 Traditional A-star algorithm

The A-star algorithm, originally proposed by Hart et al[10], is a graph-based search algorithm that finds optimal paths by using an admissible heuristic function to prioritize promising regions. Unlike breadth-first search or Dijkstra's algorithm, A-star significantly reduces the number of expanded nodes, enabling faster convergence to the optimal solution.

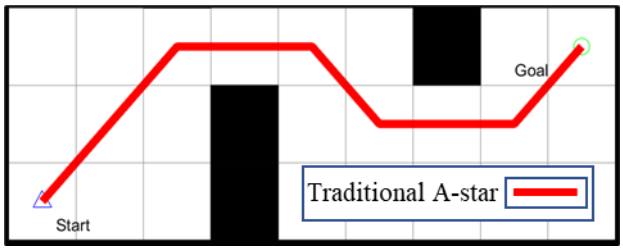


Fig. 1 Traditional A-star algorithm

As shown in Fig. 1, the A-star algorithm incrementally builds a tree from the start node to the goal. In each iteration, it selects the node with the lowest total cost from the OpenList, directing the search towards the most promising regions. The total cost at any node n is defined as:

$$f(n) = g(n) + h(n) \tag{1}$$

where f(n) represents the estimated minimum cost from n-th node to the goal, g(n) is the accumulated cost from the start node to the n-th node; and h(n) is the heuristic function estimating the cost to reach the target node.

The Euclidean distance is widely used in grid-based path-finding due to its simplicity and effectiveness.

$$h(n) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \tag{2}$$

where (x_1, y_1) and (x_2, y_2) are the coordinates of the current node and target node, respectively. This straight-line

distance provides a natural and efficient heuristic for guiding the search.

To ensure the A-star algorithm finds the optimal path, the heuristic function $h(n)$ must satisfy the admissibility condition:

$$h(n) \leq h^*(n) \quad (3)$$

Where $h^*(n)$ is the true minimum cost from node n to the goal. This condition guarantees that the heuristic never overestimates the actual cost, ensuring that no potentially better path is overlooked.

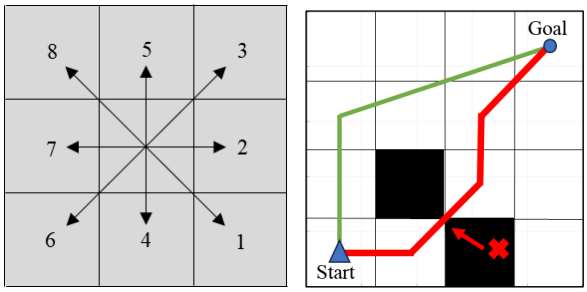
The Traditional A-star Algorithm is summarized in Table 1.

Table 1 Traditional A-star Algorithm

1. Initialization
- Add the start node to the open list (OpenList).
2. Loop Until Goal Found or OpenList is Empty
- a. Select Current Node
- Choose the node with the smallest f(n) = g(n) + h(n) from the OpenList as the current node.
- b. Check 8-Neighbor Nodes
- i. Skip Conditions
- If the node is an obstacle or in the closed list (CloseList), skip it.
- ii. New Node Processing
- If the node is not in the OpenList:
- Compute g(n), h(n), f(n) and add it to OpenList.
- iii. Existing Node Optimization
- If the node is already in OpenList:
- If the new path has a smaller g(n), update its parent node and cost value.
- c. Mark Completion
- Move the current node to the CloseList.
3. Goal Found
- Backtrack from the goal node to reconstruct the path.
4. No Path Available
- If OpenList is empty and the goal is not found, return "No Path".

2.2 Improved A-star algorithm

Although the traditional A-star algorithm possesses advantages such as high computational efficiency and global optimality in path planning.



1) 8-Connection Grid Expansion　　2) Example of an Infeasible Path

Fig. 2 Node expansion strategy of traditional A-star

Its inherent 8-neighborhood expansion mechanism, as shown in Fig. 2-1, leads to several limitations in practical applications. These include excessive path turning angles and suboptimal path planning in obstacle-adjacent regions. As illustrated in Fig. 2-2, the algorithm may generate paths that follow diagonals too closely to obstacles or even unreasonably attempt to pass through narrow gaps between obstacles.

To address these issues, Kaiyu Li proposed an improved A-star algorithm based on sub-node optimization, incorporating a sub-node classification mechanism and a dynamic neighborhood selection strategy to enhance path smoothness and safety[11]. This algorithm introduces a sub-node classification mechanism along with a dynamic neighborhood selection strategy. The key idea is to adjust the expansion algorithm based on the local environment:

- X-type nodes: In open areas where obstacles are sparse, the algorithm uses the standard 8-neighbor expansion, maintaining high search efficiency.
- N-type nodes: In the proximity of obstacles, the algorithm restricts expansion by disabling selected diagonal moves. This results in a 6-neighbor dynamic expansion, preventing the path from crossing obstacle corners and reducing unnecessary turns.

This dynamic selection of the expansion neighborhood not only minimizes redundant turns but also enhances both the feasibility and safety of the generated path.

After implementing the sub-node optimization algorithm, while the issue of diagonally crossing obstacle vertices has been resolved, another challenge remains: the problem of path smoothness and path length optimality persists. To optimize the path and improve its smoothness, we apply the Floyd algorithm[12] as a post-processing step to remove redundant nodes and enhance path efficiency.

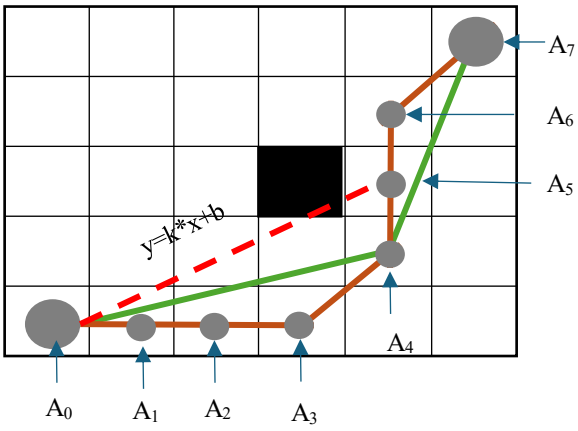


Fig. 3 Principle of path smoothness process

As shown in Fig. 3, the path $A_0 \rightarrow A_1 \rightarrow A_2 \rightarrow A_3 \rightarrow A_4 \rightarrow A_5 \rightarrow A_6 \rightarrow A_7$ is generated by the traditional A-star algorithm. This path undergoes optimization based on the following steps:

1, Collinear Node Removal: The algorithm iterates through the path to check for three or more consecutive nodes aligned in a straight line. If the direct connection between these nodes is obstacle-free, the intermediate nodes are removed, retaining only the first and last nodes. For example, if A_1, A_2, and A_3 are collinear, A_1 and A_2 would be eliminated, leaving only A_0, A_3, A_4, and A_5.

2, Direct Node Connection: Following initial simplification, the algorithm checks whether a current node can directly connect to a node further ahead while maintaining safe obstacle clearance. If a direct path exceeding safety thresholds exists, intermediate nodes are bypassed. For instance, if a safe direct path exists between A_3 and A_6, node A_4 would be removed.

By eliminating unnecessary nodes, this algorithm smooths the path, reduces turning points, and enhances the feasibility and efficiency of the navigation process. The final improved path retains only essential key nodes, ensuring a concise, efficient route while avoiding unnecessary detours around obstacles.

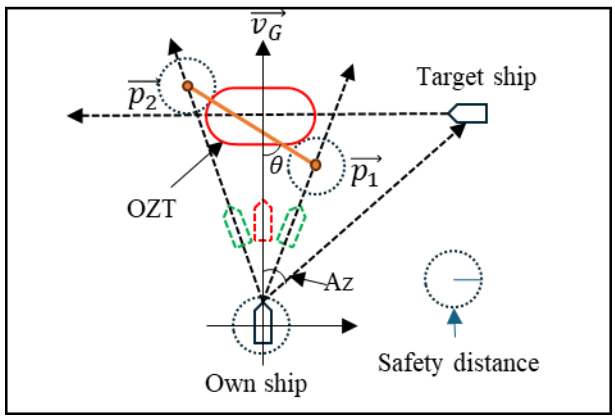


Fig. 4 Definition of OZT

2.3 OZT method

The OZT method has been widely used for collision risk assessment in maritime navigation and has evolved through various definitions. This method can identify danger zones and predict the risk areas of a target ship as shown in Fig. 4. In this research, it is defined as a capsule-shaped area and is also used to calculate evasion points.

The collision course C_O, calculated using Eq. (4), represents the trajectory of the own ship that may lead to a potential collision with the target ship in the future. This collision course can be equivalently represented by the vectors $\overrightarrow{p_1}$, $\overrightarrow{p_2}$

$$C_O = Az \pm \alpha - arcsin\left(\frac{V_T}{V_O} sin\left(Az \pm \alpha - C_T\right)\right) \quad (4)$$

Here, C_O is the collision course of the own ship, V_O and V_T are the velocities of the own ship and the target ship, Az denotes the relative bearing of the target ship from the own ship, and C_T is the course of the target ship.

The parameter α is introduced in Eq. (5):

$$\alpha = arcsin\left(\frac{SD}{d}\right) \quad (5)$$

In this equation, SD represents the safety distance, and d is the distance between the own ship and the target ship.

To assess collision risk, the relative motion of the target ship with respect to the own ship is analyzed when the own ship follows the collision course C_O. The relative motion components along the X and Y axes are calculated as:

$$\begin{cases} \Delta X = V_T \, sin \, C_T - V_O \, sin \, C_O \\ \Delta Y = V_T cos \; C_T - V_O cos \; C_O \end{cases} \quad (6)$$

Using these components, the relative speed V_R and the relative course C_R are determined as:

$$\begin{cases} V_R = \sqrt{\Delta X^2 + \Delta Y^2} \\ C_R = arctan\left(\frac{\Delta X}{\Delta Y}\right) \end{cases} \quad (7)$$

Here, V_R represents the relative speed, and C_R is the relative course of the target ship with respect to the own ship.

The minimum distance between the ships, the DCPA, and the time required to reach this point, the TCPA, are given by:

$$\begin{cases} DCPA = d|sin(C_R - Az + \pi)| \\ TCPA = \dfrac{dcos(C_R - Az + \pi)}{V_R} \end{cases} \quad (8)$$

2.4 Collision avoidance Decision-making system

After establishing the OZT, to apply COLREGs, we need to determine direction using the cross-product method. The COLREGs were established by the International Maritime Organization (IMO), prevent ship collisions. As illustrated in Fig.5, key rules governing encounter assessment and collision avoidance include:

Rule 13 (Overtaking): Vessels approaching from >22.5° abaft beam must give way by altering course to starboard or port.

Rule 14 (Head-on Situation): Vessels meeting on reciprocal courses must both alter the course to starboard, passing port-to-port.

Rule 15 (Crossing Situation): When vessels cross paths, the one with the other on its starboard side must give way by altering course to starboard and passing astern.

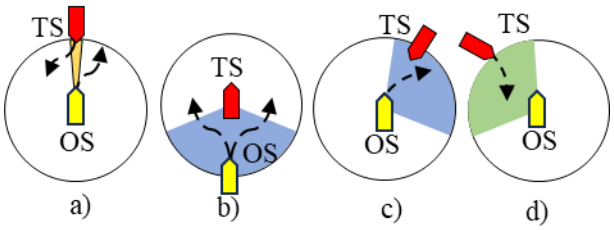


Fig. 5 Collision avoidance in four situations

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These rules provide the foundation for our collision avoidance route planner. This system is divided into three modules.

1, Risk monitoring module: Uses the OZT method to detect obstacle zones of target vessels and calculate collision points on both port and starboard sides.

2, Collision avoidance Decision-making system (CADS): Determines the encounter situation through the relative positions of the target vessel and own vessel, then makes avoidance decisions for the own vessel's evasive actions.

3, Path planning module: Imports obstacle zones and collision points calculated by OZT, selects the starboard-side evasion point as an intermediate node for the A-star algorithm when avoidance is necessary, thereby implementing direction selection and executing path planning that complies with COLREGs.

To determine the avoidance direction, we employ the vector cross-product method. As shown in Fig. 4, First, we define the vector from the own ship to the goal as $\overrightarrow{V_G}$. The vector $\overrightarrow{V_C}$ between two collision points $\overrightarrow{p_1}$, $\overrightarrow{p_2}$ is calculated as:

$$\overrightarrow{V_C} = \overrightarrow{p_1} - \overrightarrow{p_2} \tag{9}$$

We use vector cross-product for directional determination, θ is the angle between the two vectors.

$$\overrightarrow{V_G} \times \overrightarrow{V_C} = \left|\left|\overrightarrow{V_G}\right|\right| \cdot \left|\left|\overrightarrow{V_C}\right|\right| \cdot sin\theta \tag{10}$$

By analyzing the sign of the cross-product result, we can determine the right and left avoidance points. And we can use this point as an intermediate target point and input it into the A-star algorithm to generate a path that complies with COLREGs.

3. SIMULATIONS

To evaluate the performance of the improved technique and verify the feasibility of the algorithm, an algorithm validity verification experiment and an algorithm comparison verification experiment were conducted. Both experiments were carried out on the Matlab R2023a simulation platform, on a workstation featuring an AMD Ryzen 5 560H processor, 16GB DDR4 memory.

Three comparative experiment environments were constructed. By setting up different environments, the performance of the proposed improved planning algorithm was analyzed and verified in terms of path length, path smoothness, running time, and COLREGs compliance capability.

3.1 Comparison of Traditional and Improved A-star Algorithms

To evaluate the effectiveness of the improved A-star algorithm in terms of path smoothness, pathfinding efficiency, and safety, simulations were conducted on a 20×20 grid map. In this environment, "Δ" represents the starting point, and "○" represents the goal.

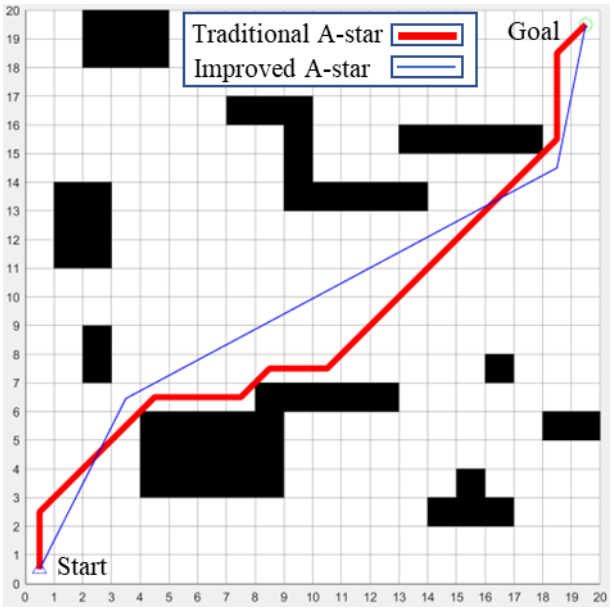


Fig. 6 Path comparison

In Fig. 6, the red line corresponds to the path planned by the traditional A-star algorithm, which collides with obstacles at coordinates (4,6), (8,7) and (18,15). In contrast, the blue line shows the path generated by the improved A-star algorithm, which employs a dynamic sub-node expansion method that avoids tangential contact with obstacles, ensuring safer obstacle avoidance and reducing the number of turning points. Table 2 presents a data comparison between these two algorithms.

Table 2 Performance comparison

Algorithm	Path Length	Turning points	Expanded Nodes
Traditional A-star	29.80	7	195
Improved A-star	28.79	2	132
Improvement	3.39%	71.43%	32.31%

3.2 Reliability of COLREGs-Compliant A-star Algorithm

The text describes an evaluation of an A-star based algorithm for compliance with COLREGs. The notation in the initial state space of the own ship is defined as: $(x_0, y_0, \theta_0, v_0, r_0) = (25.5,5,0,6,3)$.

In this five-tuple, x_0 and y_0 represent the initial coordinates of the own ship on the 50×50 grid map. θ_0 denotes the initial heading angle of the ship, measured in degrees. v_0 indicates the initial velocity of the ship, measured in grid units per time step. r_0 represents the radius of the ship, which defines its safety boundary on the grid. All experiments use goal points located near position (25, 48).

In this simulation, 4 scenarios were designed: head-on,

overtaking, starboard-side crossing, and port-side crossing. Table 3 shows the initial parameter settings of the target ships in the four scenarios.

Table 3 Parameter setting for Target ships

	x	y	θ	v	r
Head on	25	40	180	3.3	4
Overtaking	25	15	0	9	4
starboard	45	25	270	5	4

Fig. 7, 8, 9 and 10 correspond to path results of the traditional A-star algorithm and improved A-star algorithm in different situations. In the images, the blue triangle represents the own ship, while the red shape represents the target ship. In front of the red target ship, the red capsule-shaped area indicates its OZT area. The black grid cells represent the cells occupied by the target ship. Two green circles mark the avoidance points calculated by the OZT method. When the system detects a risk, the Decision-making system will make an avoidance decision. If avoidance is necessary, an avoidance point will be calculated as an intermediate node, shown as a purple circle.

In the current experiment, for each experimental scenario, the computation times for both the traditional A-star algorithm and the improved A-star algorithm are identical. The respective times are 0.0127 seconds, 0.0146 seconds, 0.0304 seconds, and 0.0082 seconds.

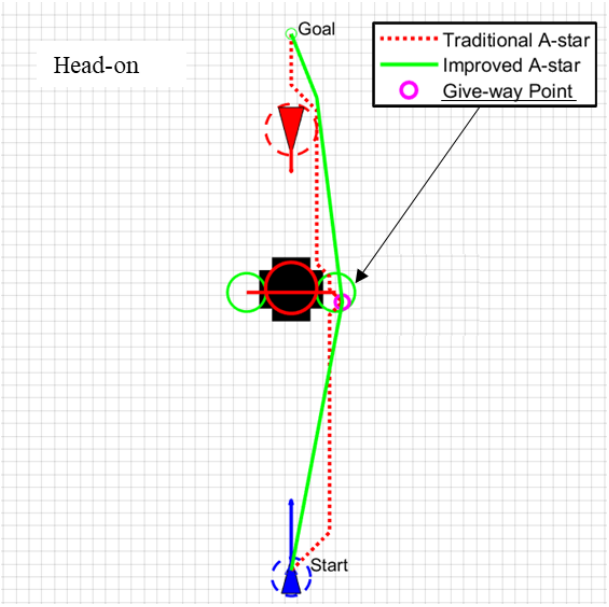


Fig. 7 Head on scenario

Table 4 Performance comparison (Head on)

Algorithm	Path Length	Turning points	Expanded Nodes
Traditional A-star	45.31	8	308
Improved A-star	42.88	2	218
Improvement	5.36%	75%	29.22%

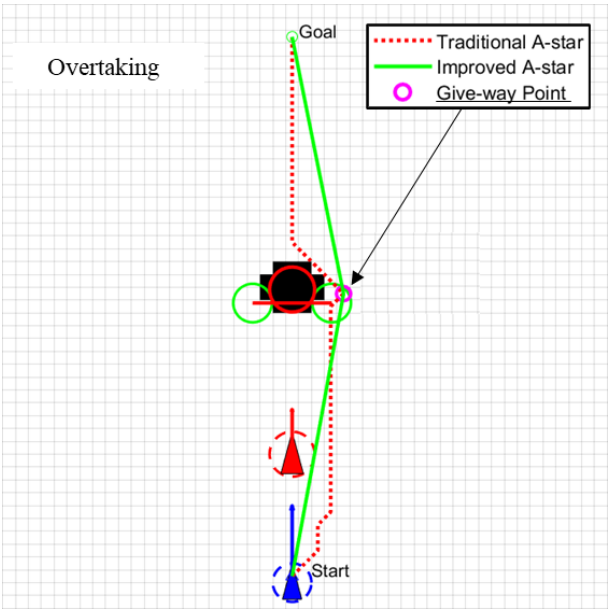


Fig. 8 Over taking scenario

Table 5 Performance comparison (Overtaking)

Algorithm	Path Length	Turning points	Expanded Nodes
Traditional A-star	45.31	6	308
Improved A-star	42.76	1	235
Improvement	5.63%	83.33%	23.70%

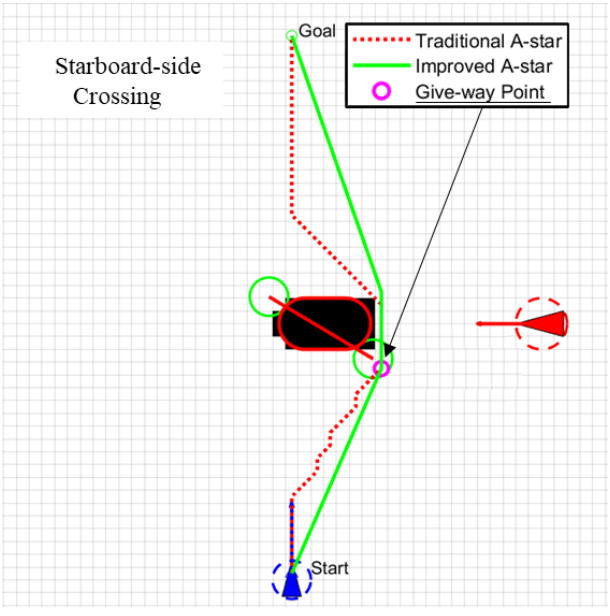


Fig. 9 Starboard-side scenario

Table 6 Performance comparison (Starboard side)

Algorithm	Path Length	Turning points	Expanded Nodes
Traditional A-star	47.80	10	371
Improved A-star	44.57	2	372
Improvement	6.76%	80%	-0.27%

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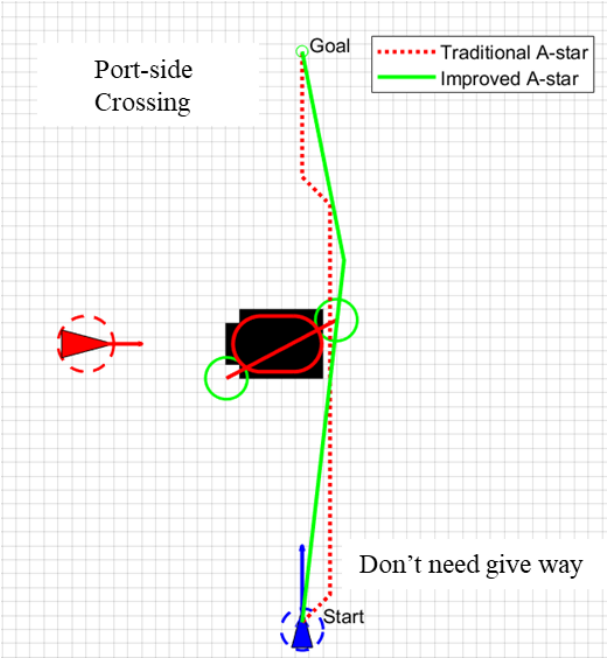


Fig. 10 Port-side scenario

Table 7 Performance comparison (Port side)

Algorithm	Path Length	Turning points	Expanded Nodes
Traditional A-star	42.66	3	272
Improved A-star	41.47	1	248
Improvement	2.79%	66.67%	8.82%

Table 4, 5, 6, and 7 show the details of the algorithm's operation. The Expanded Nodes represent the number of child nodes expanded during the algorithm's execution; fewer nodes indicate higher operational efficiency. Finally, the Improvement row details the optimization percentage in terms of path length, number of turning points, and expanded nodes.

3.3 Simulation and Validation in Tokyo Bay

3.3.1 Simulation setting and Safety Distance

A maritime algorithm evaluation was conducted through simulation experiments based on the Tokyo Bay ferry route, comparing the impact of collision avoidance decisions on traditional and improved A-star algorithms. The simulation used real route environment parameters and assessed algorithm performance in starboard crossing and head on encounter scenarios.

As shown in Fig. 11, the ferry route connecting Kurihama and Kanaya Ports spans about 11.5 kilometers with a 40-minute sailing time. This route creates crossing encounters with vessels entering the Uraga Suido Traffic Route. Despite the Tokyo Wan Vessel Traffic Service Center's advanced monitoring systems, significant collision risks remain due to high traffic density and vessel diversity. Therefore, developing efficient collision avoidance algorithms is essential for improving navigational safety in this busy waterway.

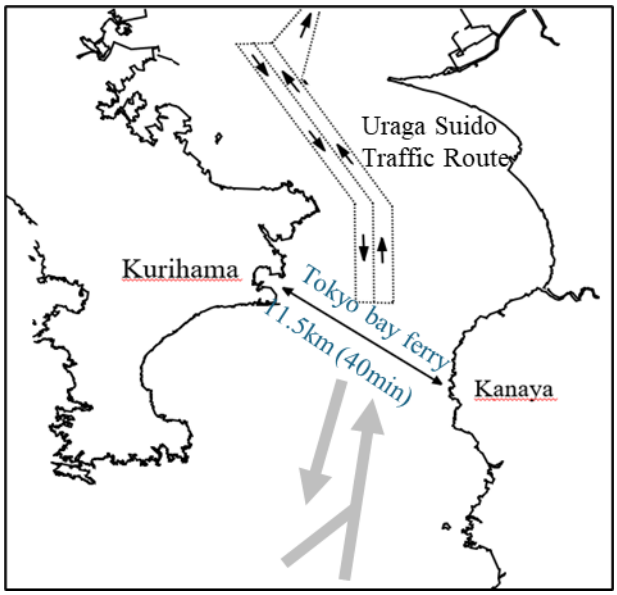


Fig. 11 Tokyo-wan ferry

Fig. 12 presents a photograph of the ferry in Tokyo Bay, providing a visual understanding of the subject in discussion. Furthermore, Table 8 details the ferry's specific parameters.



Fig. 12 Kanayamaru

Table 8 Tokyo Bay Ferry Information

Attribute	Kanaya maru	Shirahamaru
GrossTonnage (GT)	3, 580	3, 351
Speed (knots)	13	13
Length(m)	79.0	79.1

SD is one of the most critical parameters in calculating the OZT. Its determination is primarily based on the length of the own ship, the length of the target ship, and the safety coefficient.

However, because the subject of this study is the Tokyo Bay ferry, and the traffic conditions in its operating area are complex, a larger safety distance is adopted to ensure safety. According to Fukuto et al.[5], the SD is dynamically defined as the greater value between 0.07 NM and 1.1 times the own ship's length. This threshold is derived from field observations and simulator-based collision avoidance studies. Therefore, in this study, safety distance is defined by the following equation, with ship lengths originally measured in meters converted to nautical miles

$(1\ NM = 1852m)$:

$$SD = max\left\{\frac{L_{own} + L_{target}}{2 * 1852}, 0.07NM\right\} \quad (11)$$

where L_{own} denotes the length of the ship of interest, and L_{target} denotes the length of the target ship.

For the Tokyo Bay ferries under study, the minimum safety distance is set to 0.07 NM (approximately 129.64 m). When the target ship's length exceeds 180.18 m (for Kanayamaru) or 180.28 m (for Shirahamaru), a larger safety distance will be applied. This approach ensures that the required minimum collision avoidance space is maintained while accommodating vessels of different sizes, thereby avoiding unnecessary detours while ensuring safe navigation.

3.3.2 Map Construction and Discretization

Constructing a grid map is crucial in implementing a path planning algorithm. The area is divided into a 100×100 grid, with the upper-left coordinate at (139.68, 35.28) and the lower-right coordinate at (139.88, 35.12). Each grid cell represents approximately 181.93 meters east-west and 178.11 meters north-south. Although slight differences exist in grid dimensions due to geographical coordinates and Mercator projection characteristics, these discrepancies do not significantly affect the overall performance and results of the path planning algorithm. The divided map is shown on Fig. 13.

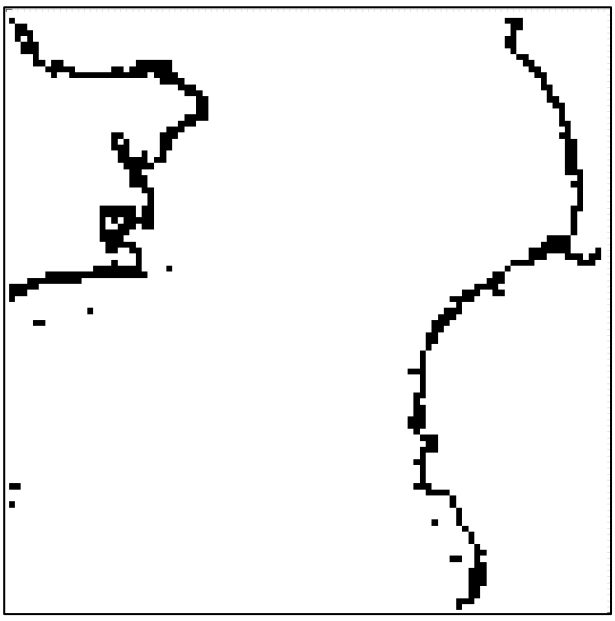


Fig. 13 Grid Map of the Uraga Suido Traffic Route Terminal Area

The process of creating a grid map from map data is a common technique used in path-planning algorithms as it allows for the efficient representation and processing of spatial information.

3.3.3 Case Studies and Performance Analysis

To evaluate the effectiveness of the proposed algorithm, a simulation was conducted using AIS data from January 1, 2018, in Tokyo Bay. The experimental data is shown in the table below.

Table 8 Experiment case setting

	Case 1	Case 2
Time	07:38	13:31
Own ship	KANAYAMARU	SHIRAHAMAMARU
Target ships	5	3
Case	Crossing	Head on

The experimental results are shown in Fig. 14 and 15, which display four different results represented by different line styles. These include the traditional A-star algorithm, the improved A-star algorithm, and each with their respective paths under CADS conditions. The triangular 'Start' symbol represents the current position of the Tokyo Bay ferry. The green circular marker indicates the ferry's destination point. The own ship's heading line is shown in blue, while other vessels' heading lines are shown in black. The heading lines predict vessel positions after 6 minutes, allowing intuitive understanding of vessel speed and direction through line length. The starting point of each heading line represents the vessel position, with circle diameters indicating vessel lengths. The red capsule-shaped areas along the heading lines represent the OZT. Black areas indicate occupied grid cells.

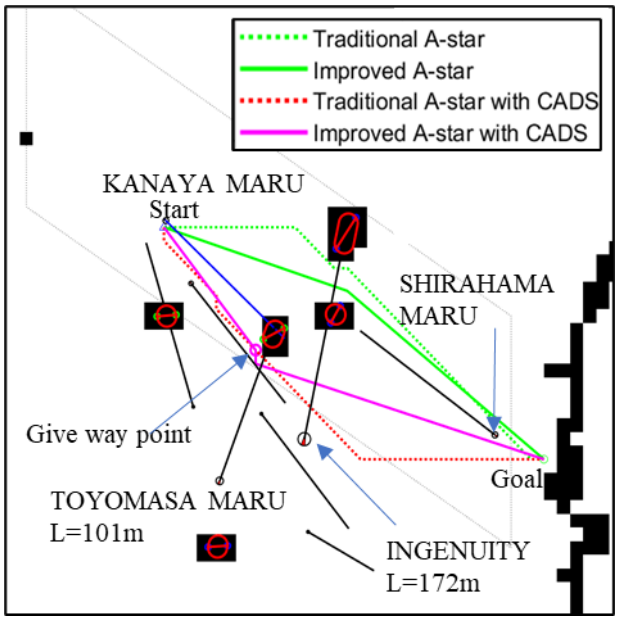


Fig. 14 Starboard-side crossing situation

Table 9 Performance comparison (Starboard side)

Algorithm	Path Length	Turning points	Expanded Nodes
Traditional A-star	37.21	4	264
Improved A-star	35.49	2	268
Improvement	4.63%	50%	-1.52%

The names and lengths of vessels with collision risks are also labeled in the images. For smaller vessels, setting

appropriate SD ensures reasonable avoidance space. Tables 9 and 10 show the calculation data and comparative analysis of the scenarios with CADS.

The computation time comparison shows that the improved A-star algorithm with CADS required 0.0218 seconds, which is 57.76% longer than the traditional A-star algorithm with CADS at 0.0138 seconds.

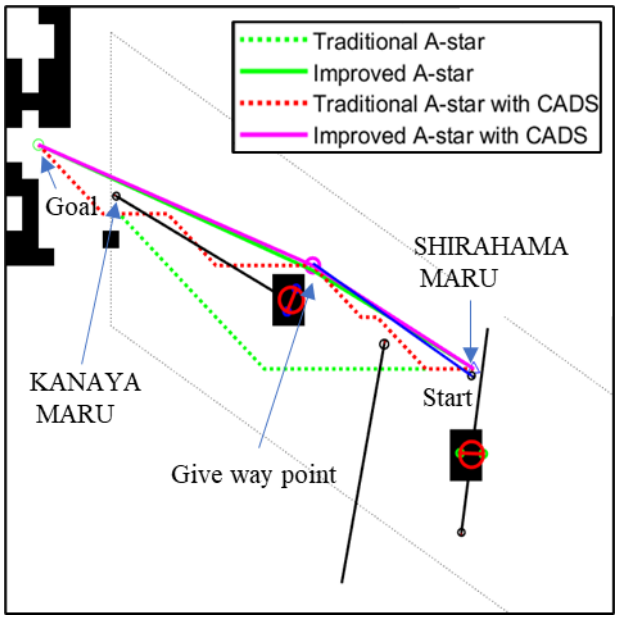


Fig. 15 Head on situation

Table 10 Performance Comparison (Head on)

Algorithm	Path Length	Turning points	Expanded Nodes
Traditional A-star	33.38	7	233
Improved A-star	31.05	2	174
Improvement	6.98%	71.43%	25.32%

Regarding computation time performance, both the traditional A-star and improved A-star algorithms with two-segment planning showed nearly identical processing times at 0.0109 seconds, with the improved version showing a negligible increase of just 0.41%.

4. DISCUSSION

The improved A-star algorithm developed in this research demonstrates clear advantages in ship collision avoidance path planning.

The improved A-star algorithm eliminates diagonal obstacle traversal risks while maintaining efficient computation across different grid resolutions (20×20, 50×50, and 100×100). Even in the most complex environment, the 0.008-second increase in processing time remains well below the 0.05-second decision threshold required for maritime applications. This improved A-star algorithm is well-suited for autonomous and semi-autonomous vessels. For fully autonomous systems, its real-time performance and COLREGs compliance provide reliable collision avoidance without human intervention, particularly valuable in busy areas like Tokyo Bay.

Path planning quality is a key performance indicator for collision avoidance algorithms. Our improved algorithm shows significant advantages in typical maritime encounters. Quantitative data reveals an average path length reduction of 5.92% across head-on, overtaking, and crossing scenarios, while path smoothness dramatically improved with an 79.44% reduction in turning points. Compared to Lee's research[13], our algorithm addresses the issue of excessive turning points in traditional A-star algorithms that result in less smooth paths, making it more suitable for large vessel navigation requirements.

The CADS system successfully integrates the improved A-star algorithm with the OZT method, effectively handling three typical encounter situations (head-on, overtaking, and starboard-side crossing). However, for port-side crossing encounters, such as vessels approaching from port, maritime management intervention is still required. The system cannot completely replace human decision-making and needs coordination from VTS and communication between crew members. In these situations, the two collision points calculated by OZT can also serve as intermediate nodes for path planning, assisting crew members in making avoidance judgments.

This improved A-star algorithm suits both autonomous and semi-autonomous vessels. For fully autonomous systems, it provides reliable, COLREGs-compliant collision avoidance without human intervention—particularly valuable in busy areas like Tokyo Bay. For semi-autonomous vessels, it offers decision support that complements human operators, providing recommended routes while allowing adjustments—ideal for the current transitional phase in maritime technology. However, limitations exist in extreme conditions where higher grid resolutions strain computational resources, and hardware dependencies challenge vessels with limited computing capabilities.

Future research will focus on three key areas. First, enhancing the system's adaptability to complex maritime environments by integrating wind, wave, and current factors into the OZT calculation model. Second, expanding autonomous decision-making capabilities for challenging scenarios, especially port-side crossing encounters, through machine learning algorithms that analyze navigation history and expert behaviors. Third, optimizing computational efficiency by implementing the algorithm in C++ and exploring parallel processing techniques to support higher resolution grid environments without sacrificing real-time performance. These improvements will advance the system toward more comprehensive autonomous navigation while ensuring strict COLREGs compliance.

5. CONCLUSION

This study developed an improved A-star algorithm with

OZT for ship collision avoidance. Experiments show it outperforms traditional A-star algorithms in three key aspects:

1. Enhanced safety through dynamic node expansion without sacrificing computation time, effectively preventing obstacle traversal risks.
2. Better path quality with reduced length and fewer turning points, generating smoother navigation paths.
3. Successful handling of various encounter situations (head-on, overtaking, and crossing) while strictly complying with COLREGs when integrated with the CADS system.

Limitations include poor adaptability to actual sea conditions, potential for computational optimization, and continued need for human oversight in complex scenarios. Future work will incorporate sea condition uncertainties, improve algorithm efficiency, and enhance autonomous decision-making capabilities for real maritime applications.

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